Trust in a Maritime Benchmark: Selection Bias, Identity Leakage, and Reproducibility in Semi-Supervised Vessel-Trajectory Classification

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***Abstract*—Semi-supervised vessel-trajectory classification from Automatic Identification System (AIS) data is a representative big-data analytics task: terabytes of noisy, multi-structured maritime broadcasts are filtered, encoded, and classified. A widely cited model, SSL-VTC, reports that adding vessel *static* features (length, width, draft) lifts four-class accuracy from 71.4% to 92.2%. We reproduce the pipeline on the same public source (nationwide MarineCadastre AIS, 2019) and audit the evaluation as a data-quality and trust problem. We find three issues. (1) *Selection bias*: the benchmark's class distribution is only reproducible if vessels lacking complete static fields are dropped: a silent filter driven by *draft*, which is missing-not-at-random (MNAR), deleting $\sim 92\%$ of fishing and $\sim 66\%$ of passenger trajectories. The published model collapses from 91.2 to 23.4 macro-F1 on the excluded vessels and scores only 64.3 on the realistic population, whereas retraining on all vessels recovers it to 88.1. (2) *Identity leakage*: the temporal split places 80.9% of test trajectories' vessels in training; we build a vessel-disjoint protocol and show the static gain does *not* collapse (transfers $\sim 91\%$), a clean negative control. (3) *Reproducibility*: the 92.2% headline depends on an undisclosed encoding hyperparameter, and the semi-supervised gain is marginal (≤ 0.7 pt) and collapses at 5% labels. We confirm the MNAR mechanism cross-region on Danish AIS, and propose a corrected, fully-disclosed evaluation protocol; all code, configs, and split indices are released.**

***Keywords*-trust in data analytics; selection bias; missing not at random; reproducibility; semi-supervised learning; vessel trajectory classification; AIS; data quality.**

I. Introduction

AIS broadcasts let coastal authorities monitor vessel traffic at scale; classifying a trajectory's *vessel type* (fishing, passenger, cargo, tanker) supports applications such as illegal-fishing detection and traffic management. SSL-VTC [1] advanced this task with a semi-supervised variational autoencoder and, notably, the integration of *static* vessel attributes (length/width/draft) alongside kinematics (position/speed/course), reporting a large accuracy gain.

We set out to reproduce and extend SSL-VTC and instead found that its *evaluation*, not its core idea, needs scrutiny. The static-feature claim is broadly *true* (static information is class-typical and even transfers to unseen vessels), but three properties of the benchmark make the reported numbers cleaner and more favorable than a realistic deployment warrants. Each property is a data-trust failure of the kind that recurs across analytics pipelines: a silent filtering step, leakage across a train/test boundary, and an undisclosed preprocessing hyperparameter, echoing long-standing warnings about dataset bias [2]. This is a re-evaluation and reproducibility study; the contribution is a corrected, disclosed protocol and a set of measurements that separate genuine signal from evaluation artifacts.

Contributions. (1) We identify and quantify a *static-completeness selection bias* (MNAR on draft) and show it materially changes which vessels are evaluated and how the published model behaves on the rest. (2) We provide a *vessel-identity-leakage audit* with a reusable vessel-disjoint protocol, showing the leakage is benign for the static-feature conclusion. (3) We document a *reproducibility gap*: the headline accuracy is governed by an undisclosed encoding hyperparameter, and the semi-supervised component is fragile. (4) We replicate the bias mechanism cross-region on Danish AIS and release code, configs, split indices, and a corrected evaluation protocol.

II. Background and Data

AIS and trajectories. Ships broadcast periodic messages; each carries a unique vessel id (MMSI). A *trajectory* is a fixed-length sequence of one vessel's messages (160 messages after the paper's extraction recipe). *Kinematic* features (LAT, LON, SOG, COG) vary over time; *static* features (LEN, WID, DRA) are constant per vessel. Following prior AIS work [3][4], each attribute is discretized into one-hot bins and concatenated ("seven-hot" encoding); the per-attribute *bin count* is a hyperparameter central to Section V. We report accuracy and *macro-F1* (mean over classes, exposing rare-class weakness) given heavy class imbalance. The setup is four classes, a temporal split (train Jan–Apr / val May / test Jun), a CNN with the seven-hot classifier, and an M2 semi-supervised VAE [5].

Data and reproduction. Our source is MarineCadastre daily AIS, 2019, Jan–Jun, U.S./Canada/Mexico coastal coverage (181 daily files, ~ 142 GB decompressed) [6]. We follow the paper's extraction (trajectory division on > 2 h gaps; ≥ 160 messages; ≥ 6 h span; abnormal-SOG removal; sample to 160) and build two cohorts: **complete**, vessels with all of LEN/WID/DRA, 133,492 trajectories; and **inclusive**, all vessels with missing static imputed, 260,543 trajectories. Faithful reproduction required matching the static seven-hot resolution (Section V); with fine bins our complete-cohort model reaches 91.6% vs the paper's 92.2%. All numbers below are the mean over three seeds. Fig. 1 plots sampled trajectories by class for both regions.

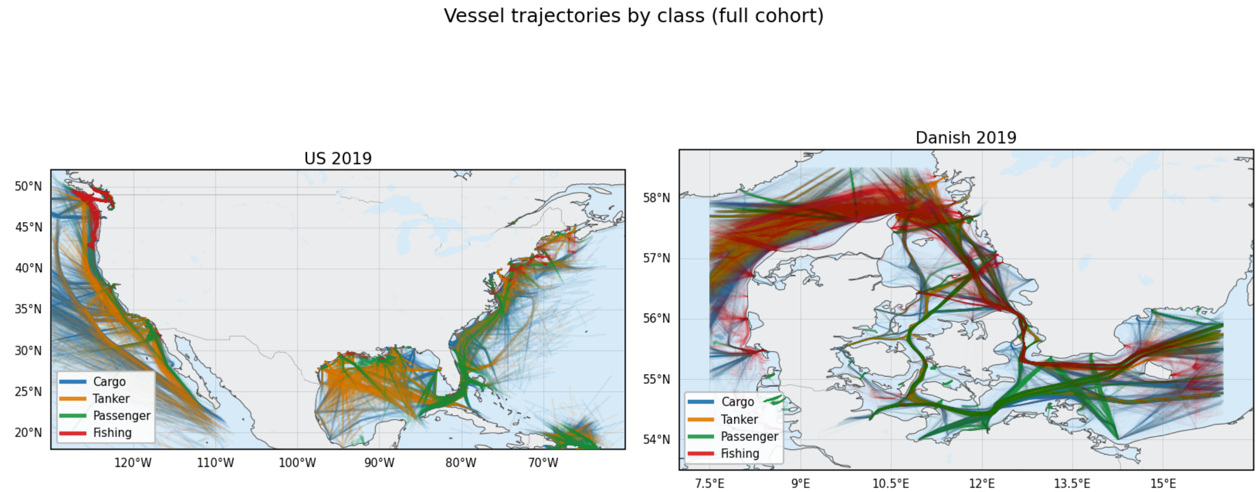

Figure 1. Sampled vessel trajectories colored by class. Left: the US MarineCadastre 2019 cohort (US/Canada/Mexico coastal coverage). Right: the Danish Maritime Authority cohort (Baltic and North Sea) used for cross-region external validity. Both show the four target classes (cargo, tanker, passenger, fishing).

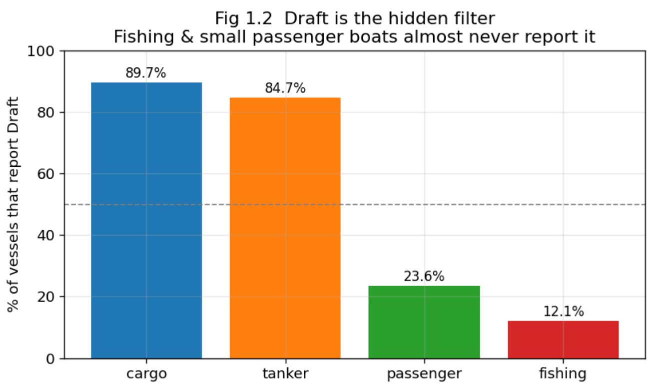

Figure 2. Static-field reporting rates by class. Length/width are near-universal, but *draft* is reported by only 12.1% of fishing and 23.6% of passenger vessels versus ∼85–90% for cargo/tanker, missingness correlated with the label (MNAR).

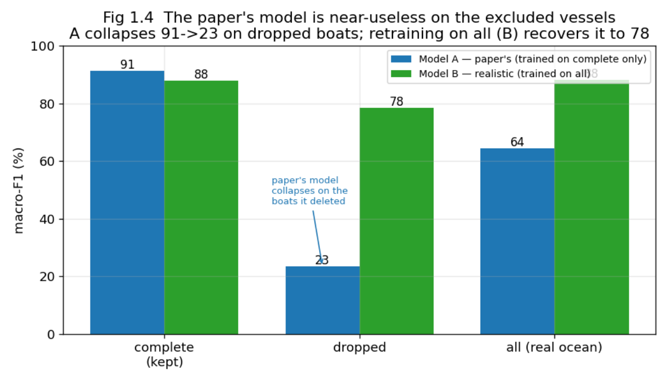

Figure 3. Macro-F1 of the published-style model (A, trained on the static-complete slice) versus a realistic model (B, trained on all vessels) across three populations. A collapses on the excluded vessels (91.2→23.4) and scores 64.3 on the real ocean; B recovers them to 88.1.

III. Finding 1: Static-Completeness Selection Bias

The benchmark's class mix is not the real ocean. Over the full training data (210.8 M messages, 12,852 vessels) raw class shares are roughly balanced, yet the benchmark is cargo-dominant: fishing falls from a raw 29.9% to 2.0% in the benchmark, while cargo rises from 27.2% to 53.8%.

Mechanism: draft is MNAR. Length and width are widely reported; *draft* is not, and its missingness correlates with vessel type (the label), the textbook definition of MNAR [7][8]. Fishing vessels report draft at 12.1% and passenger at 23.6%, versus cargo 89.7% and tanker 84.7% (Fig. 2). Requiring all three static fields reproduces the paper's class mix almost exactly (cargo 52.3 vs 53.8, fishing 3.6 vs 2.0), strong evidence the benchmark applied this filter, likely inadvertently, since the method *needs* static fields.

The filter deletes the operational target. Requiring complete static fields halves the data (260,543 → 133,492) with wildly uneven trajectory-level survival: cargo 91.4%, tanker 83.6%, passenger 34.5%, fishing 8.3%. The benchmark thus excludes ∼92% of fishing trajectories, the class central to illegal-fishing applications.

Consequence: the published model collapses on the excluded vessels. We trained the paper-strength model two ways and scored each on three populations (macro-F1; Fig. 3). Model A (trained on complete) scores 91.2 on kept vessels but collapses to 23.4 on dropped vessels and 64.3 on the real ocean; passenger and tanker recall fall to ∼0 because A relies on real size while dropped vessels carry only imputed size.[1] Model B (trained on all vessels) recovers dropped to 78.4 and scores 88.1 overall, so the failure is a protocol artifact, not a broken method. The paper's full M2 VAE collapses identically (22.6 on dropped), so the generative regularizer provides no robustness to the excluded population.

[1] A is evaluated under its own training normalization, as in deployment; the collapse is identical under matched normalization, ruling out a normalization artifact.

TABLE I. PER-CLASS RECALL (REALISTIC MODEL): KINEMATIC-ONLY VS ADDING STATIC SIZE.

Class	Motion only	+ Size	Size boost
Fishing (target)	86.8	91.5	+4.7
Passenger	82.6	89.5	+6.9
Cargo	74.4	88.9	+14.5
Tanker	61.0	83.1	+22.1

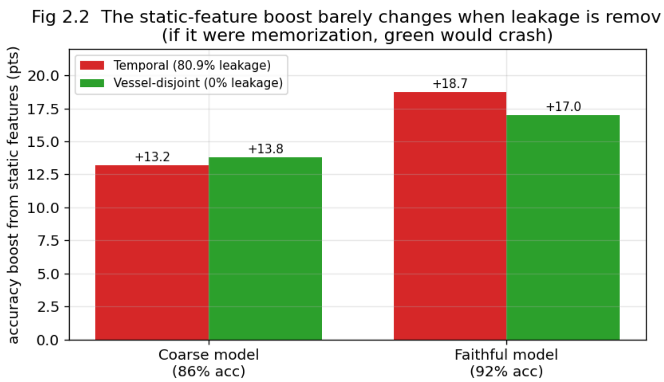

Figure 4. Static-feature gain under the leaky temporal split versus a vessel-disjoint split (0% MMSI overlap). The gain transfers ($\sim$91%), confirming the conclusion is not an artifact of identity leakage.

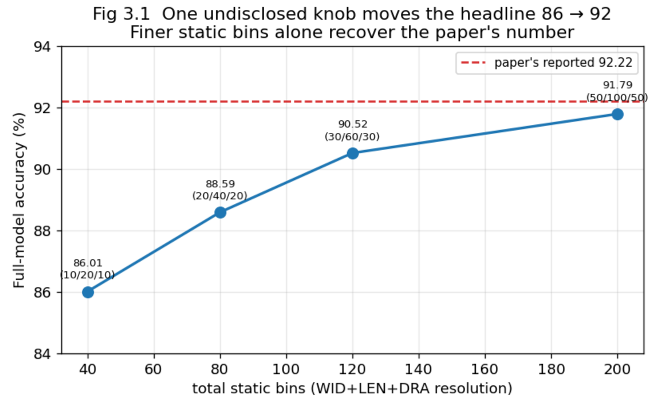

Figure 5. Full-model accuracy versus static seven-hot bin resolution (kinematic bins fixed). The undisclosed bin count alone moves accuracy 86.0$\rightarrow$91.8, closing nearly the entire gap to the reported 92.2%.

TABLE II. DRAFT-REPORTING RATE BY CLASS, US VS DANISH AIS (CROSS-REGION).

Class	US 2019	Danish 2019
Cargo	89.7%	97.8%
Tanker	84.7%	98.9%
Passenger	23.6%	77.6%
Fishing	12.1%	47.5%

The static benefit barely reaches the fishing target. Per-class recall of the realistic model, with-size vs kinematic-only (Table I), shows the size feature lifts large vessels (tanker +22.1, cargo +14.5) but barely the fishing target (+4.7), which is already at 86.8% from motion alone. The size benefit is real but accrues to large vessels; the operational fishing class gains little and is well-classified from kinematics.

IV. FINDING 2: VESSEL-IDENTITY LEAKAGE (BENIGN)

Static features are constant per vessel, an identity fingerprint. The temporal split leaks identity: 80.9% of test trajectories belong to a vessel also in training (74.6% vessel overlap); a vessel-disjoint split (partition MMSIs, stratified by class) gives 0%. We re-ran the static ablation under both protocols on a faithful 91.6% model. The static gain (Full $-$ kinematic-only) does *not* collapse: it moves from +18.7 on the leaky temporal split to +17.0 on the vessel-disjoint split (Fig. 4). About 91% of the static benefit transfers to unseen vessels, full-model accuracy drops only $\sim$2 pts, and fishing recall stays high. The leakage is real but does *not* inflate the static-feature result: a rigorous negative control. A small $\sim$9% identity component appears only at fine encoding resolution; the dominant signal is class-typical.

V. FINDING 3: REPRODUCIBILITY

The headline rides an undisclosed hyperparameter. We reproduced everything from the paper yet plateaued at 86.0% vs the reported 92.2%. The cause is the unspecified static seven-hot bin resolution. A controlled sweep (kinematic bins fixed) moves Full accuracy from 86.0 (40 bins) to 88.6 (80), 90.5 (120), and 91.8 (200) with no other change (Fig. 5). Diagnostics rule out optimizer, scheduler, and checkpoint selection: both cosine and constant learning-rate schedules cap at $\sim$86 at every epoch. The headline is therefore not reproducible from the paper as written.

The semi-supervised component is fragile. Reproducing SSL-VTC against a labeled-only baseline on faithful data, the M2 gain is marginal (+0.7 at 20% labels, +0.2 at 40%, $-$0.0 at 60%; the paper reports a clear advantage). At the 5%-label setting the model collapses to majority-class prediction: the reconstruction ELBO swamps the classifier at extreme label scarcity, precisely where the largest semi-supervised benefit is claimed.

VI. EXTERNAL VALIDITY: DANISH AIS

To test whether MNAR draft-missingness is US-specific, we replicated the mechanism check on Danish Maritime Authority open AIS (May 2019, 2 M messages, 1,464 unique vessels; a different region, fleet mix, and regulator) [9] (Table II). The rank order is identical: fishing reports draft at the lowest rate in both datasets, cargo and tanker at the highest. Absolute rates differ (Danish fishing vessels are more professionally regulated), but the structural gap (fishing $\ll$ cargo/tanker) is preserved. The MNAR mechanism is a property of AIS broadly, not a US-data artifact.

VII. A CORRECTED EVALUATION PROTOCOL

We recommend: (1) *keep static-incomplete vessels* and impute missing fields instead of deleting them (Model B recovers the excluded population); (2) *report per-class recall and macro-F1* so weak rare-class performance is visible; (3) *use vessel-disjoint splits* (no MMSI in both train and test) to remove identity leakage; (4) *disclose all preprocessing hyperparameters*, especially encoding bin resolution, or use a learned encoder to remove the knob; (5) *evaluate on the inclusive population by default*, reporting the static-complete subset only as a clearly-labeled secondary slice.

VIII. Limitations and Conclusion

We cannot prove the original filter was intentional; we show it is the unique filter reproducing the paper's distribution (cargo/tanker within $\sim$1 pt). Our data vintage differs slightly (133k vs the paper's 115k trajectories) though shapes match. Findings 1, 2, and 3.1 use the supervised CNN seven-hot classifier, exactly the model the static tables and 92.2% headline are reported on, while Finding 3.2 and the VAE collapse check use the full M2 autoencoder; the corrected protocol transfers to any model.

SSL-VTC's central idea, that static features help vessel-type classification, is sound and even transfers to unseen vessels. But its benchmark silently selects a static-reporting, cargo-heavy subpopulation on which the published model collapses when deployed realistically, leaks vessel identity (though benignly), and reports a headline that depends on an undisclosed hyperparameter. The science survives; the evaluation needed fixing. All code, configs, split indices, and figure-generation scripts are released to support reproduction.

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